

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 1)

10 divisions on the main scale of a Vernier calliper coincide with 11 divisions on the Vernier scale. If each division on the main scale is of 5 units, the least count of the instrument is :

- (1) $\frac{1}{2}$
- (2) $\frac{10}{11}$
- (3) $\frac{50}{11}$
- (4) $\frac{5}{11}$

Q2 - 2024 (27 Jan Shift 1)

Identify the physical quantity that cannot be measured using spherometer :

- (1) Radius of curvature of concave surface
- (2) Specific rotation of liquids
- (3) Thickness of thin plates
- (4) Radius of curvature of convex surface

Q3 - 2024 (27 Jan Shift 2)

Given below are two statements: one is labelled as Assertion(A) and the other is labelled as Reason (R).

Assertion (A) : In Vernier calliper if positive zero error exists, then while taking measurements, the reading taken will be more than the actual reading.

Reason (R) : The zero error in Vernier Calliper might have happened due to manufacturing defect or due to rough handling.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

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Q4 - 2024 (30 Jan Shift 1)

Young's modulus of material of a wire of length ' L ' and cross-sectional area A is Y . If the length of the wire is doubled and cross-sectional area is halved then Young's modulus will be :

- (1) $\frac{Y}{4}$
- (2) $4Y$
- (3) Y
- (4) $2Y$

Q5 - 2024 (30 Jan Shift 2)

If 50 Vernier divisions are equal to 49 main scale divisions of a travelling microscope and on smallest reading of main scale is 0.5 mm, the Vernier constant of travelling microscope is:

- (1) 0.1 mm
- (2) 0.1 cm
- (3) 0.01 cm
- (4) 0.01 mm

Q6 - 2024 (31 Jan Shift 2)

The measured value of the length of a simple pendulum is 20 cm with 2 mm accuracy. The time for 50 oscillations was measured to be 40 seconds with 1 second resolution. From these measurements, the accuracy in the measurement of acceleration due to gravity is $N\%$. The value of N is:

- (1) 4
- (2) 8
- (3) 6
- (4) 5

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Answer Key

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Solutions

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Q1

$$10\text{MSD} = 11\text{VSD}$$

$$1\text{VSD} = \frac{10}{11}\text{MSD}$$

$$\text{LC} = 1\text{MSD} - 1\text{VSD}$$

$$= 1\text{MSD} - \frac{10}{11}\text{MSD}$$

$$= \frac{1\text{MSD}}{11}$$

$$= \frac{5}{11} \text{ units}$$

Q2

Spherometer can be used to measure curvature of surface.

Q3

Assertion & Reason both are correct Theory

Q4

Young's modulus depends on the material not length and cross sectional area. So young's modulus remains same.

Q5

$$50 V + S = 49 S + S$$

$$S = 50(S - V)$$

$$.5 = 50(S - V)$$

$$S - V = \frac{0.5}{50} = \frac{1}{100} = 0.01 \text{ mm}$$

Q6

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Solutions

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$$\begin{aligned}T &= 2\pi\sqrt{\frac{\ell}{g}} \\g &= \frac{4\pi^2\ell}{T^2} \\\frac{\Delta g}{g} &= \frac{\Delta\ell}{\ell} + \frac{2\Delta T}{T} \\&= \frac{0.2}{20} + 2\left(\frac{1}{40}\right) \\&= \frac{0.3}{20}\end{aligned}$$

$$\text{Percentage change} = \frac{0.3}{20} \times 100 = 6\%$$

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